Experiment Title

Aim :- Study of LAN Topology, network Devices, and Address configuration

Problem Statement

1. Draw the network Topology:

- Identify and draw the Lan topology of & clearly label all components.

Sol → The collage campus network observed this Study consists of :-

- Layer 3 core switches in the server room.
- Managed by PoE switches
- Wireless Access point (APs)
- Eathernet Port in classroom & labaratories

Network Topology :- Hybrid hierarchical topology which combines.

(1) Tree topology
(2) Extended star topology
(3) Ring Topology (Server room)

The network hierrarchical layer

1. Core layer
2. Distribution layer
3. Access layer

- Core layer ÷ Core layer contains server room and server room have firewall, Routers, 2-L3+ Switches Patch Rack, server, inverted etc.

- Distribution layer ÷ Distribution layer 4-switches

The four switches are -

- Main building → R_7 (Room no 213, 2nd Floor)
- Main building → R_1 (Room no 301, 3rd floor)
- Main building → R_6 (Room no 113, 1st floor)
- Main building → R_2 (Room no 201, 2nd floor)

→ This 4- switches contains ÷

→ 18 Port Gigabit Easy smart switches with 16- Port (TL- SG1218MPE)

→ (SG2418XMP) → 24 Port Gigabit & 4 Port 10 GESFP+, L2+ managed switches 24 Port PoE+.

1. TP- Link TL-SG1218MPE

→ likely used for :- CCTV camera, wifi access point etc.

→ 18 Point gigabit "Easy smart switch".

→ 16 Port support PoE+

② TP- Link Omada SG3428XMP

→ Higher-end managed layer 2+ switches.

→ 24 PoE+ Part, uplinks, 10 GSFP

→ Usually act as a main distribution for each floor.

Access layer :- Access layer contain 3- switches i,e.

1. Main building R_1 (Room no - 301)
2. Main building R_4 (Room no - 05)
3. Main building R_5 (Room no 18)

→ Every switches act as access layer which helps to Provide network connectivity to all classroom, lab faculty room etc.

→ Access layer directly connect :

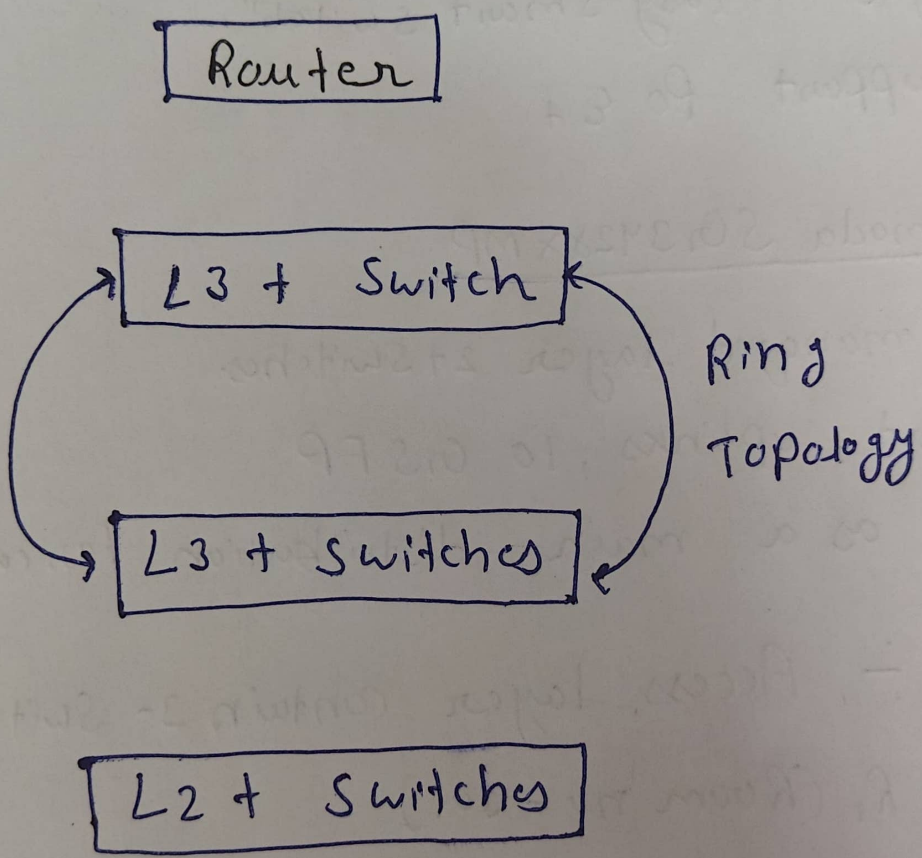
① Ethernet Port

② wireless AP

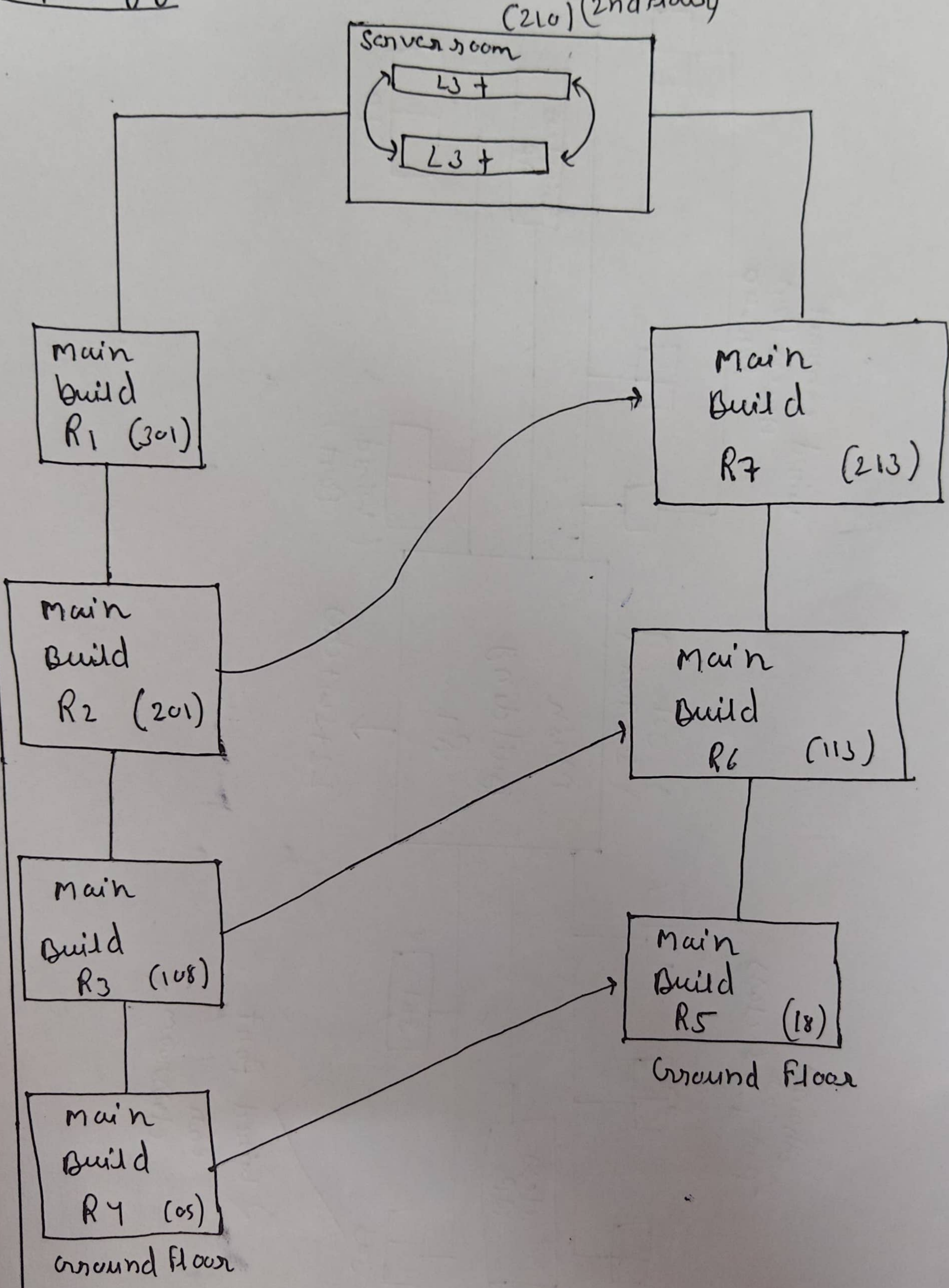
③ Lab system

4. outdoor wireless.

Server room Architecture



Topology

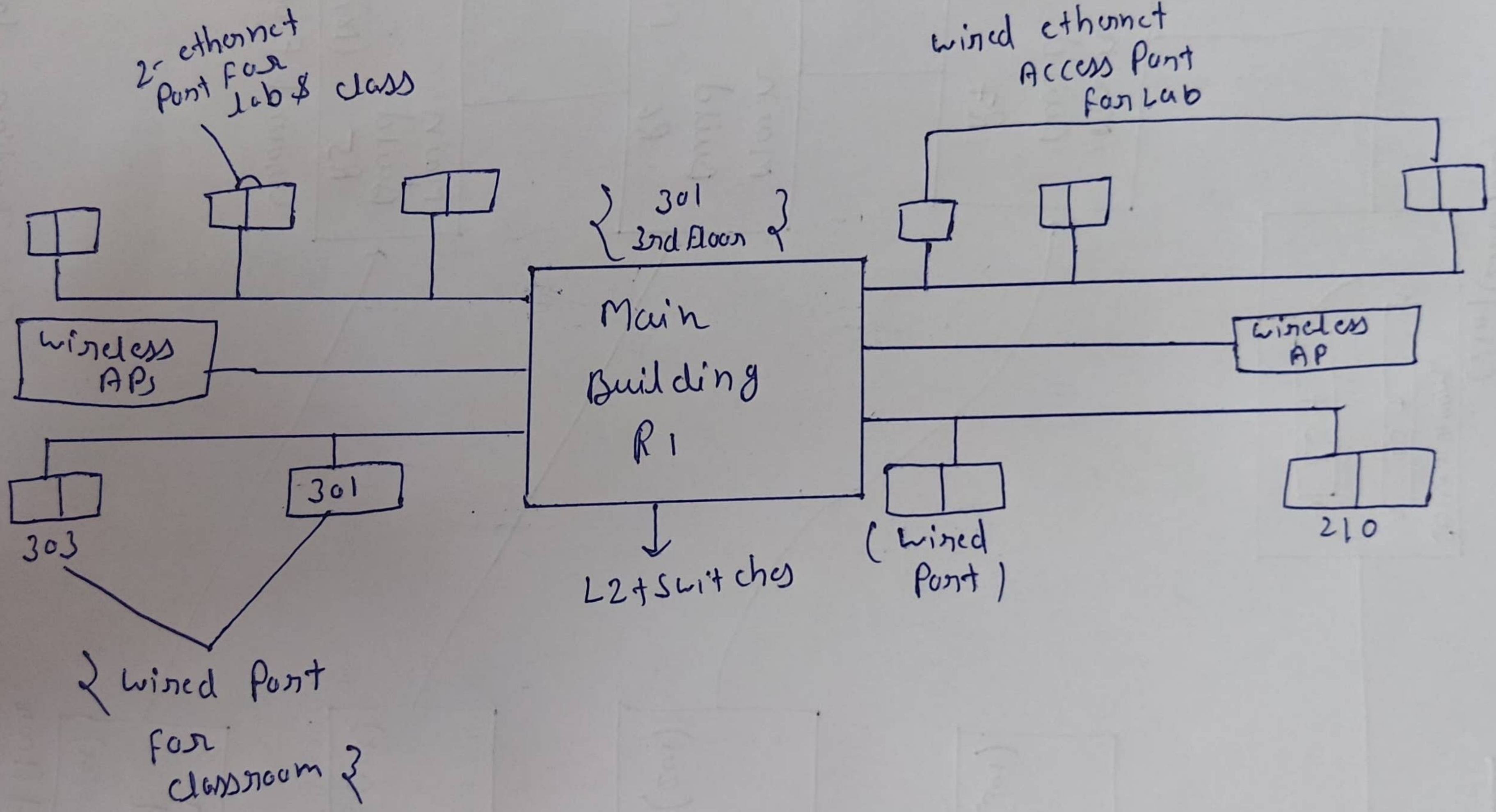


→ R1 , R7 , R2 , R6 → Distribution layer

→ Server room → core layer

→ R1, R2, R3, R4, R5 , R6, R7 → Access layer.

Access layer Switch (3rd Floor)



wired connection

Ip addres

using ipconfig

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Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\asus>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 4:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::1c58:9285:2343:ab1d%8
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter Ethernet 5:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2409:40e4:b:62c6:a33f:4036:d440:cd7b
   Temporary IPv6 Address. . . . . . : 2409:40e4:b:62c6:1d84:290:91:e9ce
   Link-local IPv6 Address . . . . . : fe80::c6fa:27dc:aeb1:7d5f%57
   IPv4 Address. . . . . . . . . . . : 10.108.64.136
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::2868:b6ff:fe40:c578%57
                                       10.108.64.135
```

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Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2409:40e4:114e:b3de:87d4:a230:a309:4989
   Temporary IPv6 Address. . . . . . : 2409:40e4:114e:b3de:7cce:2b94:8d25:be65
   Link-local IPv6 Address . . . . . : fe80::ac23:944b:300a:104d%9
   IPv4 Address. . . . . . . . . . . : 10.175.38.127
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::cccc:5cff:fe1d:bcbe%9
                                       10.175.38.181
```
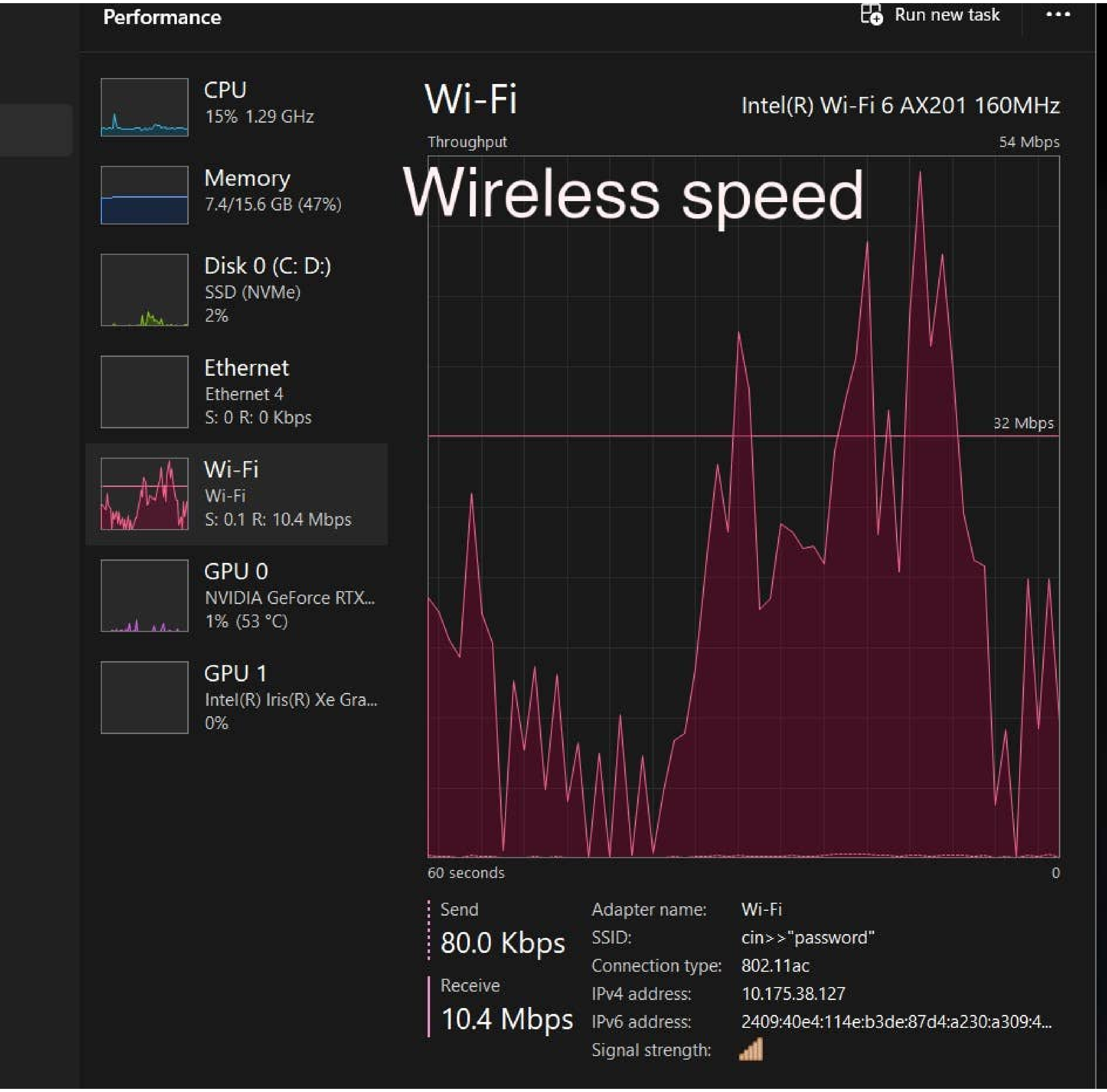

wireless connection

Ip address

using ipconfig


C:\Users\asus> getmac
Mac address
using getmac
Physical Address   Transport Name
=================== ==
E8-B0-C5-19-51-B8   \Device\Tcpip_{50CD253C-9DC9-4D65-99FC-BFEBA430418B}
CC-28-AA-17-45-B3   Media disconnected
0A-00-27-00-00-08   \Device\Tcpip_{4CBE8870-F434-419E-BD9F-99FDA7BB8E7A}
E8-B0-C5-19-51-BC   Media disconnected
00-FF-97-72-FC-FC   Media disconnected
C:\Users\asus>


Performance
Run new task
CPU
15% 1.29 GHz
Memory
7.4/15.6 GB (47%)
Disk 0 (C: D:)
SSD (NVMe)
2%
Ethernet
Ethernet 4
S: 0 R: 0 Kbps
Wi-Fi
Wi-Fi
S: 0.1 R: 10.4 Mbps
GPU 0
NVIDIA GeForce RTX...
1% (53 °C)
GPU 1
Intel(R) Iris(R) Xe Gra...
0%
Wi-Fi
Intel(R) Wi-Fi 6 AX201 160MHz
Throughput
54 Mbps
Wireless speed
32 Mbps
60 seconds
0
Send
80.0 Kbps
Receive
10.4 Mbps
Adapter name: Wi-Fi
SSID: cin>>"password"
Connection type: 802.11ac
IPv4 address: 10.175.38.127
IPv6 address: 2409:40e4:114e:b3de:87d4:a230:a309:4...
Signal strength:

Q Diff b/w wired & wireless speed?

Sol

Wired network	Wireless network
→ Faster communication	→ Slower communication
→ Stable connection	→ Affected by distance / Interference.
→ Low latency	→ High latency
→ Dedicated Physical medium	→ Shared radio medium.

Q Role of Subnet mask in network?

→ A subnet mask identifies the network Portion & host Portion of an IP address. It helps device determine whether another device belongs to the same network or diffrent network.

- It is important for:

1. Network Segment
2. Efficient routing
3. communication b/w devices.